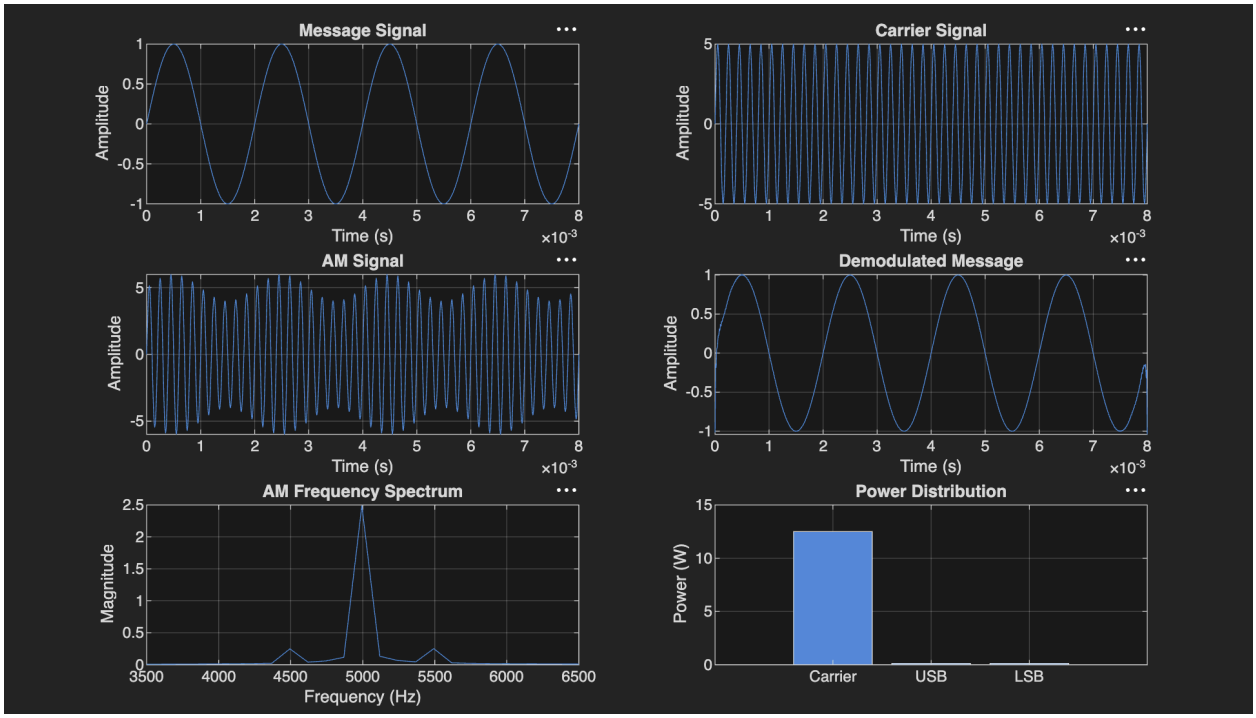
EXP NO: 2

CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

```
clc;
clear;
close all;
% Signal Parameters
Am = 1;
fm = 500;
Ac = 5;
fc = 5000;
fs = 20 * fc;
% Time Vector
t = 0:1/fs:4/fm;
% Message and Carrier Signals
m = Am * sin(2*pi*fm*t);
c = Ac * sin(2*pi*fc*t);
% Modulation Index
mu = Am/Ac;
% AM Modulation
am = Ac * (1 + mu*sin(2*pi*fm*t)) ...
  .* sin(2*pi*fc*t);
% AM Demodulation using Envelope Detection
dem = abs(hilbert(am));
dem = dem - Ac;
dem = dem / (Ac * mu);
% Frequency Spectrum
N = length(am);
f = (-floor(N/2):ceil(N/2)-1) * (fs/N);
AM_FFT = abs(fftshift(fft(am))) / N;
% Sideband Frequencies
USB = fc + fm;
LSB = fc - fm;
% Power Calculations (1-ohm load)
Pc = Ac^2/2;
Psb = (mu^2 * Pc)/4;
Ptotal = Pc + 2*Psb;
% Plot Results
figure('Name','AM Modulation','NumberTitle','off');
subplot(3,2,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(3,2,2);
plot(t,c);
```

```
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(3,2,3);
plot(t,am);
title('AM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(3,2,4);
plot(t,dem);
title('Demodulated Message');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(3,2,5);
plot(f,AM_FFT);
xlim([fc-3*fm fc+3*fm]);
title('AM Frequency Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
subplot(3,2,6);
bar([Pc Psb Psb]);
set(gca,'XTickLabel',{'Carrier','USB','LSB'});
title('Power Distribution');
ylabel('Power (W)');
grid on;
% Display Results
fprintf('Modulation Index = %.2f\\n',mu);
fprintf('USB = %d Hz\\n',USB);
fprintf('LSB = %d Hz\\n',LSB);
fprintf('Carrier Power = %.3f W\\n',Pc);
fprintf('USB Power = %.3f W\\n',Psb);
fprintf('LSB Power = %.3f W\\n',Psb);
fprintf('Total Power = %.3f W\\n',Ptotal);
```


Message Signal
Amplitude
Time (s)
×10-3
Carrier Signal
Amplitude
Time (s)
×10-3
AM Signal
Amplitude
Time (s)
×10-3
Demodulated Message
Amplitude
Time (s)
×10-3
AM Frequency Spectrum
Magnitude
Frequency (Hz)
Power Distribution
Power (W)
Carrier
USB
LSB